Review Article

JOURNAL OF ONCOLOGY PHARMACY PRACTICE

J Oncol Pharm Practice
2025, Vol. 31(2) 282–293

DOI: 10.1177/10781552241296516
journals.sagepub.com/home/opp
Sage

Pharmacist interventions in optimising opioid medication therapy in pain management for palliative care patients: A systematic review

Rajeev Shrestha[1], Sunil Shrestha[2], Ayesha Iqbal[3] and Gizem Gülpınar[4]

Abstract

Background: Opioid therapy is a critical component in managing pain in palliative care, where pharmacists' specialised expertise is crucial in ensuring quality care for patients. This systematic review aims to document available evidence on pharmacist interventions and their impact on optimising opioid therapy for pain management in palliative care patients.
Methods: We searched Medline (OVID), Embase (OVID), APA PsycINFO and Cochrane Central Register of Controlled Trials (CENTRAL) for relevant articles published from the beginning to 31st December, 2022. All original studies documenting pharmacists' intervention and impact in optimising patients receiving opioid therapy for their pain management in palliative care settings were included in this review.
Results: The database and reference search yielded to a total of 7154 studies. Out of these, only 3 studies met the eligibility criteria and were included in this study. These studies were conducted in Korea, Canada and United States. Pharmacists were involved in assessing pain, suggesting medication for pain and other symptom management, providing patient education, counselling and recommendation, assessing patient's medication effects such as adverse effects, drug interaction and duplication, and adjusting medication. Similarly, their involvement showed improvements in pain management, opioid usage and management strategies.
Conclusion: This systematic review highlights the important role of pharmacists in optimising opioid medication therapy for pain management in palliative care patients. Their contributions to palliative patient care improve pain outcomes and overall quality of life. Integrating pharmacists into palliative care teams can enhance pain management practices and provide better care for palliative patients. Further studies accompanying the robust methodologies and broader settings will validate the findings of this review.

Keywords

Opioids, pain, pharmacist, palliative care

Date received: 16 July 2024; revised: 23 September 2024; accepted: 16 October 2024

Introduction

Palliative care is a unique healthcare field aimed to enhance the quality of life of patients with life-limiting illnesses.[1,2] The fundamental element of palliative care is ensuring a comfortable and respectful experience for patients at their end stages and effectively managing their pain and other associated symptoms. Opioid therapy is a component in managing pain in palliative care, which alleviates patients from distressing symptoms while upholding their right to a pain-free life.[3]

Optimising opioid medication therapy is a vital element in palliative care, where the role of healthcare professionals, specifically pharmacists, is paramount. Pharmacists' specialised knowledge in medication selection, use, dosage adjustment and education can contribute substantially to the

[1]INF Nepal Green Pastures Hospital and Rehabilitation Centre, Pokhara, Nepal
[2]Department of Research and Academics, Kathmandu Cancer Center, Tathali, Bhaktapur, Nepal
[3]Office of Lifelong Learning and the Physician Learning Program, Faculty of Medicine and Dentistry, University of Alberta, Edmonton, Canada
[4]Department of Pharmacy Management, Faculty of Pharmacy, Gazi University, 06330, Ankara, Türkiye

Corresponding authors:
Rajeev Shrestha, INF Nepal Green Pastures Hospital and Rehabilitation Centre, Pokhara, Province Gandaki, Nepal.
Email: rajiv2stha@gmail.com

Sunil Shrestha, Department of Research and Academics, Kathmandu Cancer Center, Tathali, Bhaktapur, Nepal.
Email: sunilcresta@gmail.com

comprehensive care of palliative patients.[4,5] Particularly, the specific requirements,[6] vulnerabilities,[7] and ethical concerns[8] of palliative patients distinguish them from those of other clinical settings. Entrusted tasks provided by pharmacists encompass evaluating and enhancing medication therapy for patients facing serious illnesses throughout their care journey,[9] conducting symptom evaluations in individuals with severe conditions,[10] crafting and overseeing care plans for symptom management in such patients,[11] and devising medication deprescribing strategies[12] that align with the values, prognosis, and care goals of both the patient and their family.[13] Therefore, the significance of pharmacist interventions in palliative care surpasses the contribution of pain relief. It entails holistic, patient-centred care that resonates with the fundamental values of palliative medicine.

Although the beneficial effects of pharmacist interventions in opioid optimising in various healthcare settings have been established,[14] it is still important to understand the specific evidence in the context of palliative care patients. Despite some studies mentioning pharmacist involvement in palliative care teams,[15–18] the extent and impact of such interventions, particularly in opioid optimisation, remain unclear. This systematic review aims to address this gap by documenting available evidence on pharmacist interventions and their impact on optimising opioid therapy for pain management in palliative care patients. By synthesising the available literature, this systematic review seeks to provide a comprehensive understanding of pharmacist interventions in optimising opioid therapy for pain management in palliative care. The findings of this review are expected to contribute to the development of best practices, guidelines, and policies in palliative care, ultimately improving pain management and patient outcomes in palliative settings.

Methods

Study design

This systematic review was designed to evaluate the interventions and impact of pharmacists in optimising opioid therapy in patients receiving opioids for pain management in palliative care settings. The study followed the Cochrane Handbook for Systematic Reviews of Interventions guideline. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) was used to report this study (see supplementary file). The protocol for this review was registered in the Prospective Register of Systematic Reviews (PROSPERO) (CRD42023492148).

Search strategy and data source

Two authors (RS and SS) systematically searched literature databases. The articles were searched by four electronic databases: Medline (OVID), Embase (OVID), APA PsycINFO and Cochrane Central Register of Controlled Trials (CENTRAL). All the articles published in English up to 31st December 2022 were only included in this study. References of included studies were also reviewed to find relevant studies. The Population, Intervention, Comparison and Outcome (PICO) elements were applied to develop a study question, search strategy and study selection. Table 1 shows the details of PICO formulated for this study. The search strategy used for this study is attached in the supplementary file (see supplementary file).

Eligibility criteria

All original studies, including practice reports, documenting pharmacists' intervention and impact in optimising patients receiving opioid therapy for their pain management in palliative care were included. Studies were excluded if the patients were not specified as receiving pharmacist single or collaborative care in opioid optimisation for pain management in palliative care, if full texts could not be found, or if they were reviews or case reports.

Study selection, data extraction and analysis

After collecting studies from a database search in Covidence (an online software for systematic review), two review authors (RS and SS) independently screened the title and abstract after removing duplicates. Then, two authors (RS and AI) independently performed full-text screening and identified the eligible studies. Any discrepancies that arose were resolved by the third author (SS).

Data were initially extracted from the included studies and entered into an MS Excel spreadsheet by one author (RS). Then, another author (AI) reviewed and cross-checked extracted data and corrected if any errors were

Table 1. Framework of PICO in the systematic review.

PICO	Population, Intervention, Context, Outcomes
Population	Palliative care patients receiving opioid medications.
Intervention	Any intervention or service provided by a pharmacist alone or in collaboration with other healthcare personnel in optimising opioid therapy for palliative care patients.
Context	All kinds of palliative patient care settings - outpatient and in-patient of hospitals, home, hospice centre and community care.
Outcome	Any outcomes related to improved pain management (decreased pain intensity, symptom relief), reduced opioid-related adverse effects, increased patient and healthcare provider satisfaction, improved quality of life and cost-effectiveness.

identified. When necessary, the studies' authors were contacted to obtain additional information. A third reviewer (SS) resolved any disagreement in data extraction. The following information was extracted from the included studies based on the aim of the study: details of the included studies (author, publication year, the objective of the study, study design and setting, sample size, duration of the study, characteristics of the study population), characteristics of the intervention (details of the intervention, duration, follow up and frequency), and outcomes or impact of the intervention (information related to opioid and pain management).

As the included studies lack uniformity in pharmacist intervention, study design, outcome measurement, study settings and participants' characteristics, synthesising the findings was conducted through a summarisation and narrative review of the included studies. Pharmacist interventions were categorised into key components such as medication management, patient education, and collaborative care to systematically evaluate the intervention and understand the individual studies' intervention with meaningful comparison. A tabular presentation was followed to present the available evidence.

Risk of bias assessment

For the quality assessment of included studies, the National Heart, Lung and Blood Institute of National Institute of Health (NIH) provided a quality assessment tool for before-after (Pre-Post) studies with no control group used for our included studies.[19] The quality assessment is categorised into good, fair and poor. Quality assessment was undertaken by one reviewer (RS) and checked by a second reviewer (SS). Any disagreements were resolved by discussing with the authors' team.

Result

Study selection

Database and reference search led to 7154 number of studies. After removing duplication, 6311 studies were initially screened for titles/abstract eligibility. After that, 90 studies were taken for full-text eligibility evaluation. The bibliographies of the full-length articles were also screened. Three studies were only found eligible to be included in this review, as shown in Table 1. The most common reason for exclusion is not specifically described in opioid optimisation in palliative care settings. The search process and screening are presented in a flowchart via the PRISMA diagram in Figure 1.

Characteristics of the studies

The total number of participants included in this review is 301. These studies were conducted in Korea, Canada, and the United States. The interventions included a retrospective analysis of pharmacist interventions for pain medication management as a multidisciplinary palliative care team with a particular focus on opioid optimisation. All 3 studies were single centered studies, had adult participants and included outpatients[20,21] and in-patients admitted to the hospital.[22] Further characteristics of the included studies are provided in Table 2.

Interventions provided by pharmacist

All three included studies[20–22] involved the collaborative contribution of pharmacists with other healthcare professionals in providing care to palliative patients. All three studies[20–22] reported the pharmacist's involvement in assessing pain, suggesting medication for pain and other symptom management, providing patient education, counselling and recommendation, evaluating patient's medication effects like adverse effects, drug interaction and duplication, and adjusting medication dose, route, and forms. Ma et al.[21] study mainly reported pharmacist intervention in stopping and starting pain medications, providing follow-up visits to monitor symptoms and medication use, and reporting pain medication-related problems. Both Gagnon et al.[20] and Ma et al.[21] stated the involvement of pharmacists in assessing morphine equivalent dose calculation and medication compliance assessment. Gagnon et al.[20] study reported that pharmacists were involved in reviewing medication history, specifically opioid toxicity and supporting the homecare team to ensure efficient distribution and use of medication. Furthermore, Geum et al.[22] study found that pharmacists were mainly involved in reviewing the appropriate use of opioids and other analgesic medicine by providing education on evidence-based treatment with new analgesics to staff. Tables 3 and 4 provide the detailed and summary interventions pharmacists provided, respectively.

Pharmacist interventions impact on pain and opioids

Three studies[20–22] reported the improvement in pain outcomes with the involvement of pharmacist care. Gagnon et al. (2011)[20] reported pain improvement in 82.5%, and pain stabilised at 12.5% by week 4. Similarly, Geum et al. (2019)[22] reported a significant improvement in pain intensity ($P < 0.001$). However, Ma et al. (2016)[21] reported only some extent of improvement (On the fourth visit, severe pain: 48% to 42% and moderate pain: 20% to 18%) with significant reduction only on the third visit ($P \leq 0.005$) (Table 5).

Effects on opioid use were reported differently in the included studies. Gagnon et al. (2011)[20] reported a substantial morphine dosage reduction (average MEDD 82.1 mg/24 h at baseline to 44.5 mg/24 h at week 4). Geum et al.[22] reported the appropriate prescriptions of opioids significantly (35.9%, 35.9%, and 76.9%, on day −7, day 0, and day 7, respectively, $P < 0.001$) with improved pain assessment and active adjustment on opioid administration

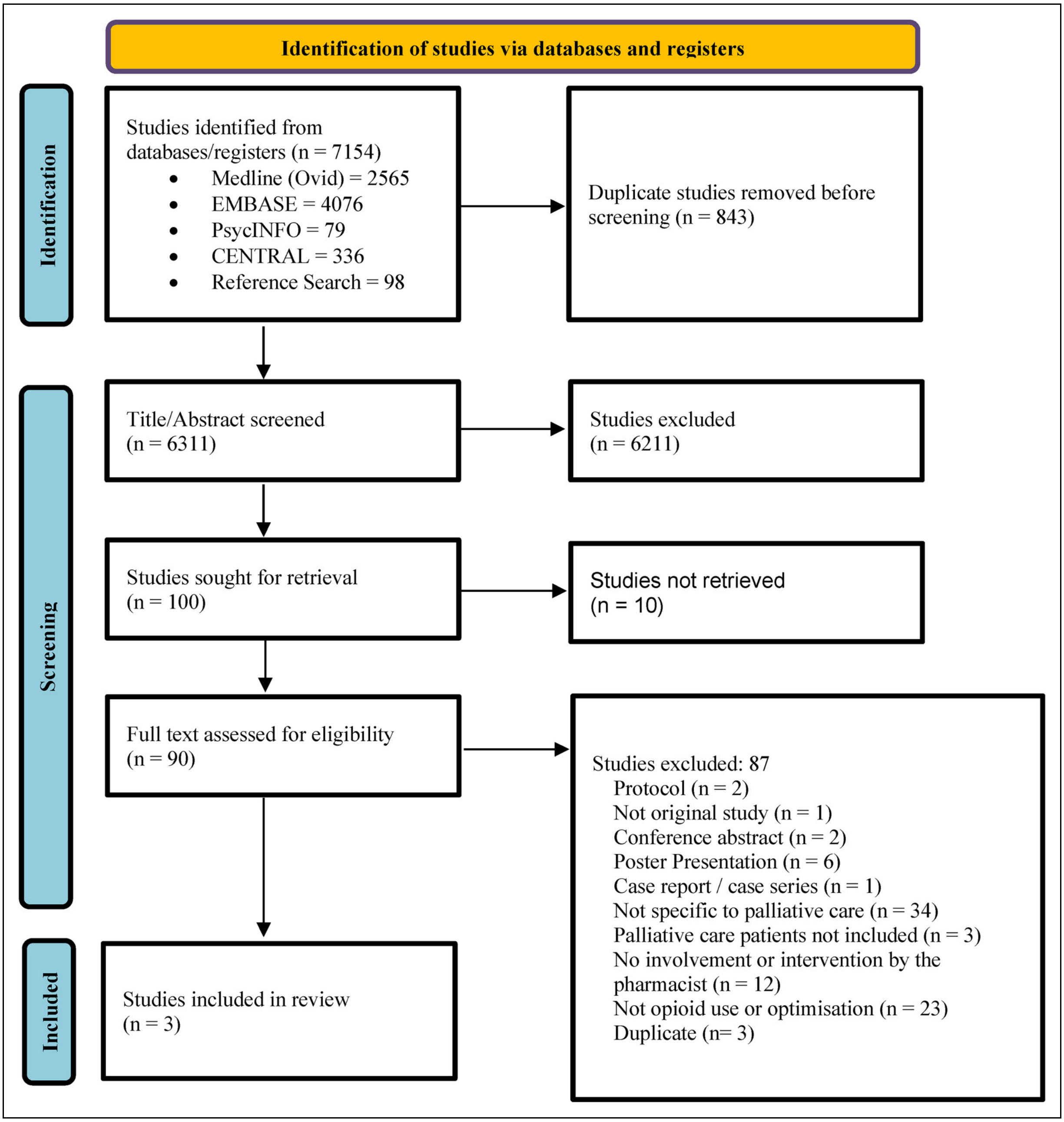

Figure 1. PRISMA flowchart of included studies

dosage. Whereas, Ma et al.[21] did not report on the appropriate dosage reduction of opioids but only on the active involvement of pharmacists in adjusting doses, frequency and changing opioids (Table 5).

Additional outcomes

All three studies reported improvement in various aspects. Gagnon et al.[20] documented a notable improvement or stabilisation of symptoms, other than pain, with pharmacist collaborative interventions: fatigue, nausea, depression, shortness of breath, anxiety, appetite, drowsiness, and the feeling of well-being. Specifically, Ma et al.[21] reported an improvement in medication problems related to patients' constipation, nausea and vomiting. Moreover, Geum et al.[22] found a significant improvement in monitoring side effects and prophylactic use of analgesics with pharmacist intervention (Table 5).

Risk of bias

Among the three included studies, two studies[20,21] were assessed as fair quality, and one[22] was of good quality (Table 2). The most common cause of bias in all included

Table 2. Characteristics of included studies

Studies	Country	Duration of the study	Study design and setting	Participants' detail	Primary disease	Patient's medication	Patients number	Quality of the study
Gagnon et al., 2011	Canada, Edmonton	Jan 1, 2007 to Dec 31, 2008	Retrospective analysis of prospectively collected data at a single outpatient palliative radiotherapy clinic	Patients aged 68.3 median aged, suffered mainly from cancer	Bone metastases (prostate (36.8%), lung (21.9%), or breast (18.4%) cancer)	Opioid and other analgesics	114 patients	Fair
Ma et al., 2015	USA, California	March 2011 to March 2012	A retrospective analysis of patient consultation data at a single outpatient palliative care centre	Patients aged 18 or above mostly suffer from cancer	Cancer, sickle cell disease (8%)	Opioid and other analgesics	N = 80 (first-time consultations and a total of 135 follow-ups - 59 on the second, 43 on the third and 33 on the fourth visit)	Fair
Geum et al., 2019	Korea	April 2014 to December 2015	Retrospective review of medical records of hospitalised patients in palliative care unit for 7 days	All patients aged 18 or above admitted to the palliative care ward for 7 or more days	Tumours in various organs and sites	Opioid and other analgesics	107 patients (49 males, mean age 58 years)	Good

Table 3. Details of intervention delivered in optimising opioid medication therapy

Studies	Intervention delivered by	Interdisciplinary team composition	Screening tools/data reported	Interventions made by Pharmacists	Duration of intervention	Assessment points or follow-ups
Gagnon et al., 2011	Clinical pharmacist	Radiation Oncologist, Registered Nurse, Nurse Practitioner, Clinical Pharmacist, Radiation Therapist, Occupational Therapist, Social Worker, and Registered Dietician	a. ESAS is an 11-point numerical rating scale- for pain and other symptoms b. KPS and PPS for functional impairment c. ECS-CP for cancer pain experience and predict the success of its management.	Clinical pharmacist interventions were medication history assessment, screening for opioid toxicity, suggesting adjuvant and supportive medications, recommending changes in dose or route of administration, identifying medication duplication, and recognising potential drug interactions and adverse reactions or therapeutic duplication, counselling and recommendation, telephone follow-up to medication changes, and initial patient assessment using various tools.	During the programme	1-week, 4-week, additional clinical pharmacist follow-up done if needed
Ma et al., 2015	Two independent prescribing pharmacists	Outpatient palliative care practice is composed of a supervising physician, a nurse practitioner, two pharmacists, and a licensed clinical social worker	a. Visual analog scale for pain scores	Under physician supervision, pharmacists assess, initiate, stop, and/ or adjust therapy to manage pain, nausea/ vomiting, and other symptoms due to lack of efficacy, adverse effects, non-adherence or missed doses, drug interactions, and duplications in therapy. Discuss with the patient to optimise medication therapy. Make documentation of recommendations for therapy and be involved in follow-up visits to monitor symptoms and medication use.	1 year	-
Geum et al., 2019	Pharmacist	Physicians, nurses, a pharmacist, a team manager, a chaplain, art therapists, social workers, a volunteer manager, and a bereavement counsellor	a. Numerical rating scale (NRS) to assess pain	Evaluating the use of analgesics, counselling patients and providing up-to-date education to staff training on evidence-based treatment with new analgesics guidelines, recommending medications, and assessing contraindications, drug interactions, and adverse effects.	1 year	Day −7, 0 (day of admission), +7

Abbreviations: Edmonton Symptom Assessment System (ESAS), Karnofsky Performance Status (KPS). Palliative Performance Scale (PPS), Edmonton Classification System for Cancer Pain (ECS-CP), Oral morphine equivalents (OMEs).

Table 4. Summary of pharmacist intervention delivered in optimising opioid medication therapy

Studies	Pain assessment	Assessment of opioid use	Morphine equivalent dose calculation	Suggesting medication for symptom management	Patient education, counselling and recommendation	Assessing medication adverse reactions, interaction, duplication	Adjustment in medication dose, route, form	Documentation of clinical recommendation	Assessment of medication compliance	Others
Gagnon et al., 2011	√	√	√	√	√	√	√		√	a
Ma et al., 2015	√		√	√	√	√	√	√	√	b
Geum et al., 2019	√	√		√	√	√	√	√		c

[a]Supporting home care team to ensure efficient distribution and use of medication, and involved in taking medication history before providing care; [b]Provides follow-up visits to monitor symptoms and medication use, reported pain medication-related problems and stop/start pain medication; [c]educating staff on new analgesic

Table 5. Pharmacist interventions effect on pain, opioid use and other outcomes

Studies	Outcome (s) in Pain	Outcome (s) in opioid	Additional outcomes
Gagnon et al. 2011	Average pain rating improved • 3.2/10 (SD 2.2) at week 1 to 2.1/10 (SD 2.4) at week 4Pain reduction • Severe pain 48.9% at baseline to 5.0% at week 4 • No pain 5.0% at baseline to 35.0% at week 4 Overall Progress by Week 4 • Pain improvement: 82.5% and stabilised at 12.5%	Average MEDD reduction • Average: 82.1 mg/24 h (Range: 0–720 mg) at baseline to average: 44.5 mg/24 h (Range: 0–400 mg) at week 4	Other symptoms improvement or stabilised from baseline to week 4 were fatigue –97.5%, nausea- 97.3%, depression-94.6%, shortness of breath-92.1%, anxiety -91.6%, appetite-86.8%, drowsiness- 80.6%, feeling of wellbeing-80.0%
Ma et al. 2016	Pain rating change • No pain: 12% on first, 5% on second, 9% on third and 6% on fourth visit • Mild pain: 20% on first, 20% on second, 16% on third and 33% on fourth visit • Moderate pain: 20% on first, 34% on second, 30% on third and 18% on fourth visit • Severe pain: 48% on first, 41% on second, 44% on third, and 42% on fourth visitNo significant ($P > 0.005$) change occurred on the second and fourth visits. Significant change ($P \leq 0.005$) was observed in third visit only.	Change in Opioid Dose: • 61% had on first, 24% on second, 21% on third and 24% on fourth visit Change in Opioid Dosing Frequency: • 23% had on first, 12% on second, 5% on third and 15% on fourth visit Start a New Long-Acting Opioid Medication: • 16% had on first, 17% on second, 7% on third and 6% on fourth visit Start a New Short-Acting Opioid Medication: • 14% had on the first, 8% on second, 5% on third and 3% on fourth visit	Constipation, nausea and vomiting-related medication problems were identified (lack of efficacy, non-adherence or dose missed, and adverse drug interaction) and mitigated with required adjustments (change dose, frequency, stop and start medication) by the pharmacist.
Geum et al. 2019	Mean Pain intensity • Significant decrease in pain intensity (3.16 ± 1.74, 4.05 ± 1.91,2.66 ± 1.23 on day -7, day 0, and day 7, respectively, $P < 0.001$) and significant negative correlation with appropriate analgesia use ($r = 0.407$, $P < 0.001$ on day -7, $r = 0.309$, $P = 0.001$ on day 0, $r = 0.241$, $P = 009$ on day 7).Pain medication use • Guideline based appropriate use of analgesic (both opioids and non-opioids) improved (35.0%, 34.2%, and 75.2% on day -7, day 0, and day 7, respectively, $P < .001$) • No significant changes in the appropriate use of NSAIDs (94.0%, 92.3%, and 96.6% on day -7, to day 0, to day 7, $P < .161$)Pain assessment • Improved in the reassessment of breakthrough pain (63.2%, 68.4% and 91.5% on day -7, day 0, and day 7, respectively, $P < .001$) • Improved in inadequate pain reassessment after opioid administration for breakthrough	Opioid use • Improved appropriate prescription of opioids significantly (35.9%, 35.9%, and 76.9%, on day -7, day 0, and day 7, respectively, $P < 0.001$) • Increased the rate of morphine intravenous push administration significantly (33.6%, 58.4%, 50.8% on day -7, day 0, and day 7, respectively, $P < 0.001$). • Decreased the rate of oxycodone immediate release tablets administration (32.8%, 29.2%, 25.4% on day -7, day 0, and day 7, respectively, $P = 0.633$). • Increased the rate of fentanyl trans-mucosal tablet administration significantly (12.1%, 8.0%, 19.2% on day -7, day 0, and day 7, respectively, $P = 0.002$). • Decreased the tramadol intravenous push administration rate significantly: 10.3%, 1.8%, and 1.5% on day -7, day 0, and day 7, respectively ($P < 0.001$).	Monitoring of side effects and prophylactic use of analgesics was significantly improved (65.0%, 65.8%, and 86.3% on day -7, day 0, and day 7, respectively ($P < .001$)

(continued)

Table 5. Continued.

Studies	Outcome (s) in Pain	Outcome (s) in opioid	Additional outcomes
	pain (27.4%, 27.4% and 8.5% on day -7, day 0, and day 7, respectively, P < .001).		

Abbreviations: Standard Deviation (SD), Morphine equivalent daily dose (MEDD), Non-steroidal Anti-inflammatory drug (NSAID).

studies[20–22] are not keeping participants blinded and measuring outcomes multiple times before and after intervention. Two studies[20,21] did not specify the pre-specified eligibility criteria and their inclusion in the study. A detailed assessment of each study's quality is attached in the supplementary file (see supplementary file).

Discussion

The systematic review examined pharmacist interventions in optimising opioid medication therapy for pain management in palliative care patients. Three included studies,[20–22] encompassing 301 participants from Korea, Canada, and the United States, provided a comprehensive overview of pharmacist-led interventions within multidisciplinary palliative care teams. This systematic review's findings demonstrated pharmacists' crucial role in pain management, opioid optimisation, and overall patient care in palliative settings.

All studies involved retrospective analyses conducted in single-centered settings, with interventions primarily focused on adult cancer patients.[20–22] Pharmacists collaborated with various healthcare professionals, including oncologists, nurses, and social workers, to deliver comprehensive care. Their interventions encompassed medication history assessments, pain evaluations, opioid dose adjustments, patient education, and the management of adverse drug reactions and interactions. Notably, Ma et al. (2015)[21] highlighted the active role of pharmacists in initiating and modifying opioid therapies based on patient needs, while Geum et al. (2019)[22] emphasised education on evidence-based analgesic use and staff training.

The systematic review highlights the significant positive impact of pharmacist interventions on pain management and opioid use in palliative care settings. Studies by Gagnon et al. (2011)[20] and Geum et al. (2019)[22] demonstrated marked improvements, with Gagnon et al. observing an 82.5% improvement in pain and a reduction in morphine equivalent daily doses (MEDD) from 82.1 mg/24 h at baseline to 44.5 mg/24 h at week 4. Similarly, Geum et al.[22] reported a significant decrease in pain intensity and improved adherence to analgesic guidelines, with appropriate opioid prescriptions increasing substantially by day 7. These studies underscore the effectiveness of pharmacist-led interventions in optimising opioid therapy, adjusting doses, and managing adverse effects, which collectively enhance pain control and patient outcomes. Conversely, Ma et al. (2015)[21] found no significant changes in pain scores by the fourth visit, though improvements were noted in managing medication-related issues such as constipation and nausea. This variability in outcomes highlights the influence of intervention specifics, patient characteristics, and study settings on the effectiveness of pharmacist interventions. The review underscores the critical role of pharmacists in palliative care, advocating for their integration into multidisciplinary teams to improve pain management and optimise opioid use.

The broader pain management strategy of pharmacist interventions showed positive impacts on other symptoms beyond pain. Gagnon et al.[20] documented improvements in fatigue, nausea, depression, and other symptoms, demonstrating the holistic benefit of pharmacist-led care. Ma et al.[21] reported enhanced management of medication-related problems, while Geum et al.[22] highlighted improvements in monitoring side effects and the prophylactic use of analgesics. These broader symptom improvements may be the efforts of pharmacists' education and counselling, such as on proper use of medications, managing potential side effects and drug interaction, strategies for managing symptoms and adherence to prescribed therapies, which were integral to the broader symptom improvements observed in the included studies.[20–22] By addressing medication-related concerns, empowering patients through personalised education, and counselling on lifestyle adjustments and non-pharmacological strategies, pharmacists helped reduce complications, improve symptom control, and enhance overall well-being.

While the primary focus of the pharmacist interventions was on pain relief and opioid optimisation, the holistic approach to medication management, which included monitoring and adjusting treatments to minimise side effects, likely played a crucial role in alleviating the other associated symptoms. However, due to the retrospective nature of these studies, no definitive causal relationships can be established between the pharmacist interventions and the improvement in non-pain symptoms. Alternatively, pharmacists' comprehensive involvement in managing the overall medication regimen underscore the broader role of pharmacists in enhancing the overall quality of life for palliative care patients. Since the current studies lack validated quality-of-life assessment tools to provide a comprehensive understanding of how pharmacist interventions influence the overall quality of life in palliative care patients, future studies should incorporate these tools.

The review identified potential biases in the included studies, primarily related to participant blinding and the lack of clear reporting on multiple outcome measurements. Two studies were assessed as fair quality,[20,21] and one a good quality,[22] indicating a need for more rigorous study designs in future research. To mitigate these issues, future studies should ensure the blinding of participants and investigators to minimise performance and detection bias and establish a clear protocol for participant inclusion and outcome measurement at multiple time points, both before and after interventions, to obtain robust data. Addressing these biases will be crucial in strengthening the evidence base for pharmacist interventions in palliative care.

Regional differences in healthcare systems, cultural attitudes towards pain management, and pharmacists' role in clinical care may have influenced the results observed in the included studies. In Korea, the healthcare system's approach to palliative care and opioid management may differ significantly from that of North America due to variations in regulations, resource availability, and cultural norms surrounding end-of-life care. Similarly, Canada and the USA, while geographically close, have distinct healthcare systems that may impact the extent and manner in which pharmacists are integrated into palliative care teams. These differences highlight the importance of context when interpreting the results of pharmacist interventions in optimising opioid therapy across different healthcare settings.

Implications for practice and policy

This review underscores the pivotal role of pharmacists within the multidisciplinary palliative care teams as a strategy to optimise opioid therapy and enhance patient outcomes. Pharmacists bring specialised expertise in medication management, crucial for the complex pharmacotherapy often required in palliative care.[23,24] Through assessing pain, adjusting opioid dosages, managing adverse effects, and providing patient education,[14,25] pharmacists can significantly contribute to pain control and opioid usage, reducing medication-related complications and enhancing overall patient well-being.

However, integrating pharmacists in established palliative care teams might be challenging in real-world practice due to limited awareness of pharmacists' potential contribution, lack of supportive policies and practice protocol, insufficient training on palliative care, resource constraints, competing priorities and lack of collaborative practices.[26–29] This is particularly challenging in developing countries.[30,31] The possible way to overcome this is to train pharmacy professionals on a comprehensive understanding of palliative care, palliative pharmacotherapy, interdisciplinary collaboration skills, patient communication and counselling skills. Practical experience through case-based learning and clinical rotations is crucial for applying theoretical knowledge. Additionally, ongoing continuing education and training in ethical and legal issues are essential to keep pharmacists updated on best practices and navigate palliative care's complexities. Similarly, developing and implementing pharmacy-integrating policies and clinical practice guidelines for the pharmacist role can ensure consistent, high-quality care across different healthcare settings. Policies should prioritise collaborative practices, where pharmacists work closely with physicians, nurses, and other healthcare professionals to deliver comprehensive palliative care. Furthermore, policymakers should consider revising healthcare delivery models and reimbursement structures to incentivise the inclusion of pharmacists in palliative care.

Recognising the vital role of pharmacists improves clinical outcomes and supports the efficient use of healthcare resources by reducing unnecessary hospital admissions and minimising the use of inappropriate medications.[32,33] By implementing these policy changes, healthcare systems can enhance the quality of palliative care, ultimately leading to better patient experiences and outcomes.

Strength and limitation

To our knowledge, this is the first study documenting pharmacist interventions and their impact on optimising opioid therapy for pain management in palliative care patients. This study provides evidence of the pharmacist's significant role in enhancing future patients' health outcomes and quality of life in palliative care settings. Several limitations of this review must be acknowledged. First, the review only included peer-reviewed published studies, excluding grey literature and unpublished research. This was intended to ensure that the included studies had undergone a rigorous peer-review process, thus maintaining the quality and reliability of the findings. However, this decision may have introduced publication bias and limited the scope of the review by omitting potentially valuable insights from non-published sources. Additionally, the review focused specifically on pharmacist-led opioid optimisation services in palliative care, resulting in the inclusion of only three studies, all of which were conducted in high-income countries. This limits the generalizability of the findings to other regions, particularly low- and middle-income countries, where healthcare practices and resources differ significantly. Another limitation is that all of the included studies were retrospective, which increases the risk of selection and recall biases that could affect the accuracy of the reported outcomes.

Furthermore, the methodologies employed in these studies lacked rigour, which diminishes the strength of the evidence. A key limitation of this systematic review is the presence of common biases in the included studies, such as the absence of blinding and unclear reporting. The lack of blinding may have introduced detection bias, potentially leading to an overestimation of the effectiveness of pharmacist interventions, as outcome assessors may have been influenced by their awareness of the intervention. Additionally, unclear reporting, particularly regarding the methods used to assess outcomes, could have caused inconsistencies in data interpretation and variations in the results. These biases limit the ability to draw definitive conclusions from the review and highlight the need for future research to adopt more rigorous study designs. Prospective studies with standardised outcome reporting are necessary to validate these findings. While this review provides valuable insights into the role of pharmacists in opioid management within palliative care, its conclusions should be interpreted with caution due to these methodological limitations.

Future research direction

To address the limitation and strengthen the findings, future research should prioritise large-scale longitudinal and multicentre studies employing rigorous methodologies to address identified biases and to evaluate effective strategies. Furthermore, comparative studies using validated outcome measures on patient outcomes, including quality of life and healthcare resource utilisation, will provide a comprehensive understanding of pharmacist interventions. Additionally, exploring the real-world integration of pharmacists into multidisciplinary teams across diverse healthcare systems and identifying the strategies to improve policies, education and training interventions will be essential for contextually adopting these findings in practice. Moreover, combining the qualitative insights from patients, caregivers, and healthcare providers with quantitative data can provide a more nuanced understanding of intervention efficacy and pharmacist skill development.

Conclusion

This systematic review highlights the potential role of pharmacists in optimising opioid medication therapy for pain management in palliative care patients. Their contributions to pain assessment, opioid dose adjustment, and patient education improve pain outcomes, which eventually enhance the overall patient's quality of life. By integrating pharmacists into palliative care teams and developing supportive policies, healthcare systems can enhance pain management practices and provide better care for palliative patients. However, further studies accompanying the robust methodologies and broader settings will validate the findings of this review.

Author contributions

RS and SS conceptualised the systematic review with AI and GG. RS and SS designed and implemented the search strategy. RS, SS, and AI screened and selected eligible studies. RS and AI performed the data extraction and analysis. All authors drafted the initial manuscript and critically reviewed and revised the manuscript. All authors reviewed and approved the final manuscript for journal submission.

Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Ethics approval and consent to participate

Not applicable.

Funding

The authors received no financial support for the research, authorship, and/or publication of this article.

ORCID iD

Rajeev Shrestha https://orcid.org/0000-0003-1822-3969

Supplemental material

Supplemental material for this article is available online.

References

1. World Health Organization. Palliative care, https://www.who.int/news-room/fact-sheets/detail/palliative-care (Published 2020, accessed 9 September 2022).
2. Australian Government Department of Health and Aged Care. What is palliative care?, https://www.health.gov.au/topics/palliative-care/about-palliative-care/what-is-palliative-care (Published October 30, 2023, accessed 20 March 2024).
3. Hodiamont F, Jünger S, Leidl R, et al. Understanding complexity - the palliative care situation as a complex adaptive system. *BMC Health Serv Res* 2019; 19: 157.
4. Wilby KJ, Mohamad AA and AlYafei SA. Evaluation of clinical pharmacy services offered for palliative care patients in Qatar. *J Pain Palliat Care Pharmacother* 2014; 28: 212–215.
5. Wilson S, Wahler R, Brown J, et al. Impact of pharmacist intervention on clinical outcomes in the palliative care setting. *Am J Hosp Palliat Med* 2011; 28: 316–320.
6. Laabar TD, Saunders C, Auret K, et al. Palliative care needs among patients with advanced illnesses in Bhutan. *BMC Palliat Care* 2021; 20: 1–10.
7. Bittencourt NdM, Duarte SdC, Marcon SS, et al. Patient safety in palliative care at the end of life from the perspective of complex thinking. *Healthcare* 2023; 11: 2030.
8. Abernethy AP, Capell WH, Aziz NM, et al. Ethical conduct of palliative care research: enhancing communication between investigators and institutional review boards. *J Pain Symptom Manage* 2014; 48: 1211–1221.
9. Krumm L, Bausewein C and Rémi C. Drug therapy safety in palliative care—pharmaceutical analysis of medication processes in palliative care. *Pharmacy* 2023; 11: 160.
10. Atayee RS, Sam AM and Edmonds KP. Patterns of palliative care pharmacist interventions and outcomes as part of inpatient palliative care consult service. *J Palliat Med* 2018; 21: 1761–1767.
11. Geiger J, Enck G, Luciani L, et al. Evolving roles of palliative care pharmacists. *J Pain Symptom Manage* 2022; 64: e357–e361.

12. Bishop LD, Rosenberg-Yunger ZRS and Dattani S. Pharmacists' perceptions of the Canadian opioid regulatory exemptions on patient care and opioid stewardship. *Can Pharm J / Rev des Pharm du Canada* 2021; 154: 394–403.
13. Bacon M, Kematick B, Suliman I, et al. Development of PharmPAL: a collaborative practice pharmacy clinic in ambulatory palliative care. *Am J Hosp Palliat Med* 2023; 40: 475–479.
14. Iqbal A, David Knaggs R, Anderson C, et al. Role of pharmacists in optimising opioid therapy for chronic non-malignant pain; a systematic review. *Res Soc Adm Pharm* 2022; 18: 2352–2366.
15. O'Connor M, Fisher C, French L, et al. Exploring the community pharmacist's role in palliative care: focusing on the person not just the prescription. *Patient Educ Couns* 2011; 83: 458–464.
16. Gilbar P and Stefaniuk K. The role of the pharmacist in palliative care: results of a survey conducted in Australia and Canada. *J Palliat Care* 2002; 18: 287–292.
17. Hussainy SY, Box M and Scholes S. Piloting the role of a pharmacist in a community palliative care multidisciplinary team: an Australian experience. *BMC Palliat Care* 2011; 10: 1–12.
18. Crul M and Oosterhof P. The oncology pharmacist as part of the palliative treatment team. *Int J Pharm Pract* 2020; 28: 92–96.
19. National Heart Lung and Blood Institute. Study quality assessment tools, https://www.nhlbi.nih.gov/health-topics/study-quality-assessment-tools (accessed 16 April 2024).
20. Gagnon L, Fairchild A, Pituskin E, et al. Optimizing pain relief in a specialized outpatient palliative radiotherapy clinic: contributions of a clinical pharmacist. *J Oncol Pharm Pract* 2012; 18: 76–83.
21. Ma JD, Tran V, Chan C, et al. Retrospective analysis of pharmacist interventions in an ambulatory palliative care practice. *J Oncol Pharm Pract* 2016; 22: 757–765.
22. Geum MJ, Ahn JH, Kim JS, et al. Interprofessional collaboration between a multidisciplinary palliative care team and the team pharmacist on pain management. *Am J Hosp Palliat Med* 2019; 36: 616–622.
23. Cortis LJ, McKinnon RA and Anderson C. Palliative care is everyone's business, including pharmacists. *Am J Pharm Educ* 2013; 77: 21.
24. Kuruvilla L, Weeks G, Eastman P, et al. Medication management for community palliative care patients and the role of a specialist palliative care pharmacist: a qualitative exploration of consumer and health care professional perspectives. *Palliat Med* 2018; 32: 1369–1377.
25. Shrestha S, Iqbal A, Teoh SL, et al. Impact of pharmacist-delivered interventions on pain-related outcomes: an umbrella review of systematic reviews and meta-analyses. *Res Soc Adm Pharm* 2024; 20: 34–51.
26. Olsen WM, Freeman C, Adewumi A, et al. A scoping review of health system guidelines for pharmacist responsibilities when dispensing opioids. *Explor Res Clin Soc Pharm* 2023; 12: 100382.
27. Hartung DM, Hall J, Haverly SN, et al. Pharmacists' role in opioid safety: a focus group investigation. *Pain Med* 2018; 19: 1799–1806.
28. Thrimawithana TR, Spence M, Lee M, et al. The role of pharmacist in community palliative care—a scoping review. *Int J Pharm Pract* 2024; 32: 194–200.
29. Offu OF, Visram S, Rathbone AP, et al. Factors influencing pharmacists' roles in preventing prescription and over-the-counter opioid misuse: a systematic review and narrative synthesis. *Int J Pharm Pract* 2024; 32: 133–145.
30. Adisa R and Anifowose AT. Pharmacists' knowledge, attitude and involvement in palliative care in selected tertiary hospitals in southwestern Nigeria. *BMC Palliat Care* 2019; 18: 107.
31. Sakeena MHF, Bennett AA and McLachlan AJ. Enhancing pharmacists' role in developing countries to overcome the challenge of antimicrobial resistance: a narrative review. *Antimicrob Resist Infect Control* 2018; 7: 63.
32. Atayee RS and Edmonds KP. A specialist palliative pharmacist enriches the interdisciplinary team: a case series discussion illustrating the entrustable professional activities for specialist hospice and palliative care pharmacists. *J Palliat Med* 2023; 26: 1755–1758.
33. Dooms M. Pharmacists are initiators in palliative care for patients with rare diseases. *Orphanet J Rare Dis* 2023; 18: –5.